

UDAYA SAI CHIKKA

AI Engineer — Conversational AI | GCP Dialogflow CX & CCAI Platform

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PROFESSIONAL SUMMARY

AI Engineer with 7+ years building and scaling conversational AI systems for customer service, IT support, and finance teams. Comfortable owning a bot end-to-end — from designing the conversation logic across different AI platforms through building the Python/FastAPI services behind it, wiring up voice and messaging channels, and deploying on Cloud. Strong track record of shortening resolution times, growing self-service adoption, and keeping releases reliable through structured testing and iteration.

SKILLS

Conversational AI Platforms	Dialogflow CX, GCP CCAI Platform, CX Agent Studio, Rasa, Microsoft Bot Framework, Botpress, IBM Watson Assistant, Voiceflow, Genesys, Twilio, IVR, CCAI, GTP, NLP, NLU, OKF, RAG, Graph RAG, CrewAI, Claude, Gemini and Open AI
Backend & Webhooks	Python, REST APIs, FastAPI, Flask, Django, OAuth2
Agent Assist	Knowledge Assist/Search, Smart Reply, AI Coach, Summarization, Sentiment Analysis
Cloud & Deployment	Google Cloud Platform, Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS
Other	JavaScript, HTML/CSS, n8n, MongoDB, Airtable, Vector Databases, Scrapy, Firestore

PROFESSIONAL EXPERIENCE

AI Engineer May 2025 – Present

Red Global, Remote, India

- Design and build Dialogflow CX agents for production contact-center deployment, shaping how conversations branch, hand off, and resolve across channels.
- Develop backend webhooks in Python and FastAPI to power the agent's logic, integrate open-source LLM, and manage prompt orchestration.
- Deploy and run webhook services on GCP Cloud Run, with CI/CD via GitHub Actions and Docker-based builds.
- Work with contact-center tooling such as Genesys and Twilio to route calls and hand off conversations to live agents.
- Test and evaluate agent behavior through unit testing and A/B testing before releases to catch regressions early.
- Build end-to-end ETL pipelines for automated bot deployment using CX Agent Studio, Agent Assist and Genesys in GCP.

Technologies: Dialogflow CX, GCP CCAI Platform, Cloud Run, Genesys, Twilio, Python, FastAPI, REST APIs, Docker

AI Consultant Jan 2025 – May 2025

Upwork, Remote, India

- Delivered a production-ready RAG-based AI system for enterprise IT support, cutting average ticket resolution time by 20% and increasing self-service rates by 15%.
- Built a Python/FastAPI tool leveraging fine-tuned CodeGen models for automated unit testing, reducing post-deployment bugs by 10% and improving edge-case coverage by 15%.
- Designed and deployed a Dialogflow CX agent for omnichannel customer support with dynamic routing, reducing live-agent transfers by 25% and improving self-service resolution by 20%.
- Built a GPT-powered system to summarize financial reports and surface trends, cutting manual analysis time by 30% and accelerating decision-making.

Technologies: Python, Dialogflow CX, FastAPI, Cloud Run, LangChain, RAG, CodeGen, GPT, REST APIs, GCP

GCP CCAI Engineer Jul 2024 – Dec 2024

Miratech, Remote, India

- Led development of next-generation chatbot and voice-bot solutions across GCP CCAI, Dialogflow CX, and Microsoft Azure Bot Framework, automating key business processes and improving customer experience.
- Integrated the LLM (playbook) feature within Dialogflow CX, improving the agent's ability to understand complex queries and deliver personalized responses.

- Built voice bots using STT, TTS, and SSML, connecting them into contact-center platforms like Genesys and Twilio for live-agent handoff.
- Collaborated with cross-functional Agile/Scrum teams to gather requirements and ship conversational features, running A/B testing to validate improvements and reduce regressions.

Technologies: Google Cloud CCAI, Dialogflow CX, Microsoft Azure, Bot Framework, Genesys, Twilio, STT/TTS/SSML, Git, CI/CD

AI Engineer Oct 2023 – Jul 2024

Vocalime Remote, Italy

- Designed and built a production-level chatbot API using a Graph RAG architecture for enhanced contextual understanding and grounded responses.
- Developed FastAPI endpoints to handle real-time requests, orchestrate knowledge-graph queries, and manage LLM interactions.
- Engineered ingestion and embedding pipelines for Confluence, SharePoint, and internal wikis into a vector store for semantic search.
- Implemented custom retrieval to extract precise context for a fine-tuned LLM, and collaborated with product/engineering to meet business objectives.

Technologies: FastAPI, Rasa, Python, Graph RAG, LangChain, Vector Databases, Confluence, SharePoint, Docker

Lead AI Engineer Oct 2021 – Sep 2023

Landslo Remote, Canada

- Engineered conversational AI applications (Python/JavaScript) integrated with client systems via REST APIs, enabling real-time data exchange and contributing to a 15% client revenue increase.
- Optimized AI solutions via structured A/B testing, achieving 25% faster response times and improved user engagement.
- Designed conversational interfaces powered by ChatGPT, integrating and fine-tuning GPT-3/ChatGPT with RAG to ground responses in real-time data.
- Led the full LLM lifecycle — data collection, preprocessing, model selection, fine-tuning, evaluation, deployment, monitoring, and maintenance.
- Built automation workflows with Python, Airtable, and n8n, and prototyped agentic AI use cases (AI SDR, real estate assistant).

Technologies: Python, Rasa, JavaScript, ChatGPT, GPT-3, RAG, REST APIs, Airtable, n8n, A/B Testing, Docker

Software Engineer May 2019 – Oct 2021

TCS, Hyderabad, India

- Developed and maintained Python web-scraping and automation scripts, replacing manual processes and improving data accuracy and reliability.
- Designed and managed Python applications and desktop tools using Django and Flask, improving operational efficiency by 30%.
- Built responsive web applications (HTML, CSS, JavaScript, Django/Flask) with interactive, real-time dashboards, consolidating data from multiple sources.
- Built 10+ chatbots using Rasa Framework for Banking, Finance and Retail customers.

Technologies: Python, Rasa, Flask, HTML, CSS, JavaScript, Scrapy, SQL, MongoDB

SELECTED ACHIEVEMENTS

- Drove a 10% monthly revenue increase for an e-commerce client by implementing a conversational AI-driven sales assistant.
- Boosted chatbot engagement by 25% and cut response times by 20% through strategic GPT-3 integration.
- Scaled chatbot infrastructure to handle a 50% surge in user traffic while maintaining responsiveness and accuracy.
- Reduced conversational error rates by 15% and support ticket volume through targeted NLU and flow tuning.
- Automated routine customer inquiries and support tasks, resulting in a 20% reduction in operational costs.
- Maintained client retention rates exceeding 90% through proactive relationship management.

EDUCATION

Electronics and Communication Engineering — Vishnu Institute of Technology, Bhimavaram (2015–2019, 87%)

Machine Learning Nanodegree — Udacity (2018)

Intermediate (MPC) — Sri Gowtami Junior College, Narsapuram (2013–2015, 98.3%)

SSC — ZP High School, Seetharamapuram (2013, 9.7 GPA)